

SHIVAM TRIPATHY

shivamtrip99@gmail.com • (412) 909-5994 • [linkedin.com/in/shivamtripathy](https://www.linkedin.com/in/shivamtripathy) • [Portfolio Website](#)

SUMMARY

Robotics Software Engineer with 4+ years of experience building and deploying real-world robotic systems including UAVs, humanoids and autonomous vehicles. Strong background in autonomy (SLAM, estimation, motion planning) and systems engineering.

EDUCATION

Carnegie Mellon University – The Robotics Institute

M.S. Robotic Systems Development | GPA: 3.87/4.0

Commencement Speaker, School of Computer Science (2024) • Swartz Entrepreneurial Fellow (2022) • TA: Systems Engineering for Robotics (2024)

Selected Coursework: ML, Robot Planning, Computer Vision, Robot Learning, Visual Learning, SLAM

May 2024

Pittsburgh, PA

MIT World Peace University

B.Tech Mechanical Engineering | GPA: 8.75/10

Special Achiever Award (2021)

May 2021

Pune, India

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, PyTorch, ROS2, Gazebo, NVIDIA Isaac Sim, MoveIt/Nav2, PX4, MAVLink, Git, OpenVINS VIO

Hardware & Sensors: RGBD Cameras, LiDAR, IMU, GPS, NVIDIA Jetson, Raspberry Pi, EO/IR Sensors

Robotics & AI: SLAM, Visual-Inertial Odometry, Motion Planning (RRT*, MPC, LQR), Kalman/Particle Filters, Imitation Learning, Reinforcement Learning, LLM-based Task Planning, Computer Vision (YOLO, CLIP)

WORK EXPERIENCE

Apellix | Robotics Software Engineer

Jun 2024 – Present

- Led autonomy stack development for power-wash and spray-painting UAVs deployed in commercial and US Army applications.
- Integrated visual localization and SLAM for GPS-denied autonomous navigation on Apellix drone platforms.
- Managed full flight software stack: state estimation, controls, state machine, collision avoidance, and failsafe systems.
- Implemented vision-based end-to-end imitation learning framework mapping raw camera input to flight control outputs.
- Built Ground Control Station app with autonomous draw-to-clean feature, enabling a SaaS business model.

CMU Robot Caregiving Lab | Graduate Research Student

Sep 2022 – Dec 2023

- Built full autonomy stack for an assistive mobile manipulator: LiDAR + stereo + IMU fusion for SLAM (AMCL), A*/DWA planning, and 3D obstacle avoidance.
- Developed cloud-integrated LLM agentic framework connecting natural language user requests to HRI, navigation, and manipulation pipelines.
- Successfully deployed robot in a Pittsburgh care facility; work featured by CMU School of Computer Science.

Symbol Robotics | Robotics Software Intern

May 2023 – Aug 2023

- First employee; built baseline robotics software stack for a 32-DoF humanoid mobile manipulator.
- Implemented RTAB-Map Visual-Inertial SLAM with dual RealSense D435i cameras for localization; integrated ROS-Nav + MoveIt! for autonomous navigation and manipulation.
- Devised long-horizon task planning agentic framework using GPT-3.5.

Asteria Aerospace | Robotics Systems Engineer

Jul 2021 – Jul 2022

- Developed target position estimation algorithm for surveillance UAVs fusing IMU, GPS, elevation maps, and YOLO detections.
- Led 360° antenna tracking system and ground control software development for multi-rotor and VTOL platforms.

DRDO / Sun Mobility | Robotics Intern / Systems Intern

Oct 2020 – Jun 2021

- DRDO: Implemented PID control and sensor filtering for gait-assistance exoskeleton robots.
- Sun Mobility: Built automation software for simultaneous multi-battery charging, achieving 1.5x faster charge times.

SELECTED PROJECTS (CMU)

Assistive Robot Autonomy: Sensor-fused SLAM + LLM planning stack deployed in a real care facility for autonomous fetch-and-retrieval tasks.

Drone MPC: Applied Model Predictive Control on a simulated drone to track moving subjects while avoiding dynamic obstacles.

CubeSat State Estimation: Extended Kalman Filter for orbital determination of a 1U CubeSat in LEO; Particle Filter for differential-drive pose estimation.

Deep RL for Nuclear Fusion: Policy-based RL to maximize plasma pressure in simulated fusion reactors while avoiding plasma tearing instabilities.